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Numerical study of the dispersion and combustion of dense layered aluminum particles driven by a central explosion

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ABSTRACT

The dispersion and combustion of dense aluminum (Al) particle layers subjected to strong central explosions are critical for understanding the energy enhancement mechanisms in novel composite explosive charges. This paper numerically investigates the entire process from initial dense packing to full dispersion and combustion using a four-way coupled Eulerian-Lagrangian method. Inter-particle collisions are resolved via a coarse-grained discrete element model, and the Al particle burning rate is evaluated by a hybrid combustion model. The particle cloud evolution, mechanical characteristics, and combustion behavior are systematically analyzed, with particular emphasis on the mechanisms governing initial perturbation formation, jet growth, and shock-induced combustion enhancement. Parametric effects of particle diameter and initial central pressure are also examined. Results reveal that heterogeneous interphase momentum transfer and alternating positive-negative gas vorticity are the primary trigger for initial perturbations at the cloud interface, which subsequently evolve into finger-like particle jets under radially dependent drag. Inter-particle collisions promote perturbation formation and jet growth, as confirmed by a collisionless simulation. Analysis of particle dynamics reveals that the cloud motion transitions from early collision-dominated to later drag-dominated behavior as dispersion proceeds. Reflected shock waves in confined spaces significantly enhance the combustion rate, increasing the peak burning rate by approximately 110% compared to open-space conditions, mainly due to the rise in oxygen concentration and secondarily to the elevated gas temperature. Smaller particles (2 μm) exhibit poorer initial dispersion and lower early-stage combustion rates, yet undergo more vigorous combustion in later stages. Increasing the initial central pressure substantially enhances early-stage combustion, although the benefit diminishes beyond 5 GPa. This study elucidates the mechanisms governing the dispersion and combustion of dense Al particle layers under strong shock loading, providing insights for the design of novel aluminized composite explosives.

Novelty and significance statement: This work provides a high-fidelity numerical investigation of the entire dispersion-to-combustion process of dense Al particle layers under gigapascal-scale central explosions, resolving the transition from packed to dilute regimes. It reveals that heterogeneous interphase momentum transfer and alternating positive-negative gas vorticity control initial cloud perturbations, which develop into finger-like jets driven by radially dependent drag, while inter-particle collisions accelerate perturbation formation and jet growth. Reflected shock waves in confined geometries enhance the peak burning rate by 110% compared to the open case, primarily through increased oxygen concentration and secondarily through elevated gas temperature. The study also uncovers a non-monotonic, stage-dependent effect of particle size on combustion efficiency and identifies a saturation threshold for initial pressure beyond which further increases yield diminishing returns. These insights establish a new mechanistic understanding of shock-induced heterogeneous combustion in novel segregated explosive architectures.

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1. Introduction

Metal particles, such as aluminum (Al), are widely employed as energy enhancers in high explosive (HE) formulations due to their high energy density and excellent thermodynamic stability [1]. Compared to conventional pure HEs, the addition of energetic metal particles substantially increases the total energy output and work capacity of explosive devices [2,3]. More importantly, the combustion time scale of metal particles is 2–3 orders of magnitude longer than the detonation time scale of explosives. This temporal asynchrony enables metal combustion to prolong energy release and enhance quasi-static overpressure, thereby extending the spatial and temporal coverage of explosion effects.

In traditional metallized explosive designs (Fig. 1(a)), metal particles are directly blended into the HE matrix [5,6]. As the detonation wave sweeps over the particles, they absorb heat and combust behind the Chapman-Jouguet (CJ) plane. However, this internal mixing configuration has inherent limitations. First, adding metal particles reduces the detonation pressure and velocity of the base explosive [7,8]. Second, excessively high metal content can cause detonation failure. More critically, the energy enhancement does not increase monotonically with metal content, exhibiting a maximum activation efficiency limit [9]. Additionally, new-generation HEs such as HNIW and DNTF have inherently low stability, and direct mixing with metal particles poses severe safety and compatibility issues [10]. These bottlenecks have driven research into novel composite charge architectures.

To overcome the drawbacks of traditional metallized explosives, a composite configuration with metal particles placed outside the explosive (Fig. 1(b)) has attracted increasing attention in recent years [11,12]. In this arrangement, the central HE charge is surrounded by an annular shell of packed metal particles. Upon detonation, the divergent shock wave from the central explosive accelerates the particles outward, inducing dispersion. The dispersed particles then ignite and combust upon interacting with shock-heated air or detonation products. Experimental studies have shown that this segregated configuration not only avoids the detrimental effects of metal particles on detonation performance but also yields higher energy output efficiency than traditional metallized explosives under certain conditions [13,14]. For instance, Xiao et al. [14] conducted confined vessel explosion experiments and found that composite charges with externally loaded Al particles exhibited a 28% increase in energy release compared with traditional aluminized explosives, with a notable enhancement in shock wave impulse. Owing to the physical separation between components, this configuration offers favorable physicochemical and detonation properties, providing new design degrees of freedom. Nevertheless, the mechanisms underlying the energy enhancement of this segregated configuration remain to be elucidated.

Understanding the working mechanism of this novel composite charge requires clarifying the dispersion dynamics of dense particle layers under explosive loading and their strong coupling with the subsequent ignition and combustion. Specifically, the strong explosion generates a divergent shock wave that first strikes the densely packed particle layer, driving its transition from a dense state to full dispersion. During this process, complex momentum and energy exchange occurs among particles, shock waves, and the gas phase, accompanied by the onset and growth of multiphase flow instabilities. The dispersed particles then mix with surrounding oxidizers and ignite, releasing chemical energy that feeds back into the flow field evolution. This entire sequence involves intricate phenomena including shock-particle interaction, interparticle collision, gas-particle mixing, and heterogeneous combustion, exhibiting pronounced multiscale and multiphysics coupling. Notably, during the transition from the initial packed state to full dispersion, the particle dynamics span dense, transitional, and dilute gas-particle flow regimes.

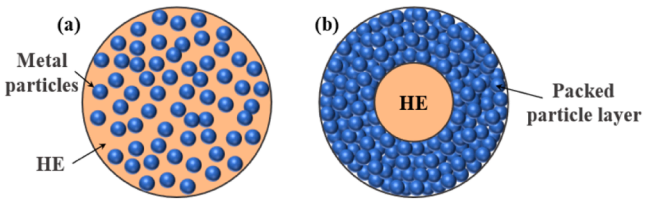


Fig. 1. Schematic diagrams of two typical configurations of metallized explosive charges [4].

Previous studies have extensively analyzed the dispersion dynamics of inert particle systems under explosive loading [15–18]. Rodriguez et al. [15] used a Hele-Shaw experimental platform to study the fragmentation and jet formation of particle rings under weak shock loading, observing that particle jets first form at the inner interface and then develop outward; they established scaling relations between jet number and both initial overpressure and particle density. Xue et al. [16] categorized the explosive dispersion behavior of granular media into four modes and linked these modes to the strength of particle-gas coupling. Koneru et al. [17] emphasized the crucial role of vorticity deposition, analogous to the Richtmyer-Meshkov instability, in guiding particles to form well-defined jets. Chiapolino and Saurel [18] found that external particle jets emerge in cylindrical and spherical explosion configurations but not in planar geometries, underscoring the importance of radial flow divergence. However, these studies focused mainly on weak shock loading (initial explosion gas pressures on the order of megapascals) and did not consider particle combustion. In practical aluminized composite explosives, the shock wave pressure from central explosive detonation can reach gigapascal levels, and Al particles undergo vigorous combustion after dispersion, resulting in physical mechanisms different from those of inert particle dispersion under weak shocks.

Experimental investigation of explosive dispersion and combustion of reactive particles faces significant challenges. The destructive explosion environment, optical opacity of particle media, and transient multiphase flows make fine-scale measurements extremely difficult. Frost et al. [19] provided photographic evidence of hydrodynamic instabilities in heterogeneous explosions but could not quantitatively characterize mixing details. Existing studies [4,20] have observed two distinct ignition modes: rapid ignition, where particles ignite almost immediately after dispersion in shock-heated air, forming an expanding luminous fireball; and delayed ignition, where particles first form a large dust cloud, followed by ignition and dust flame propagation. However, due to experimental limitations, the coupling mechanisms between particle dispersion and combustion, multidimensional effects, and the influence of reaction kinetics remain unclear.

Numerical simulation provides a powerful tool for investigating these phenomena. Two numerical approaches are commonly used. The first is the Eulerian-Eulerian (E-E) method, treating both gas and particle phases as continua. Balakrishnan et al. [21] developed an empirical two-fluid model for particle ignition and combustion induced by blast waves, but it neglects volume fraction effects and applies only to dilute conditions. Posey et al. [4] employed the E-E method to study the ignition and combustion of annular Al particle shells dispersed by surrounding TNT charges, parametrically examining the effects of layer thickness, initial packing, and particle diameter. Although the E-E method is simple and computationally efficient for the dispersed phase, its continuum assumption renders it generally unsuitable for dilute gas-particle flows and the transitions between dense and dilute regimes. Moreover, as gas-particle reactions proceed, differences in particle combustion rates increase polydispersity [22], posing significant challenges for the E-E approach.

An alternative is the Eulerian-Lagrangian (E-L) approach, which tracks the motion and combustion of individual particles in Lagrangian coordinates. The Lagrangian particle model is widely regarded as the optimal method for simulating particle dynamics and polydisperse systems. Balakrishnan and Menon [23] used the E-L method to study the flow field from a spherical TNT explosion in environments with Al

particle clouds, but that study considered dilute conditions and neglected particle collisions and volume fraction effects. In our recent work [24], we developed a four-way coupled gas-particle reactive flow numerical method applicable from dense to dilute regimes. This method uses an E-L framework to describe the evolution and interaction of the gas and particle phases, incorporating volume fraction effects and interphase coupling in the gas-phase governing equations. Particle collisions are resolved with a coarse-grained discrete element model, and a hybrid combustion model [25] is applied to evaluate the Al particle burning rate.

Thus, building on the above numerical framework and model, the present study performs full-process simulations of the dispersion and combustion of a dense Al particle layer under central explosive loading. The objectives are to elucidate the dispersion mechanisms of the dense particle layer driven by strong shock waves and the combustion characteristics of Al particles, and to further analyze the effects of reflected shocks in confined spaces, initial particle diameter, and explosion energy.

2. Physical problem and numerical methods

2.1. Physical problem

This study employs numerical simulations to investigate the dispersion, mixing with air, and subsequent combustion of dense layered Al particles driven by a diverging blast wave from a central explosion source. Owing to the geometric symmetry, only one-quarter of the domain is modeled, as illustrated in Fig. 2. The computational domain is a square of side length L, with the bottom and left boundaries defined as symmetry planes. The top and right boundaries are set as either walls or outlets to simulate confined or open-field environments, respectively. The explosion source is modeled as a hot, pressurized air pocket surrounded by an annular ring of Al particles with a particle volume fraction (α_p) of 50%. The radius of the high-pressure zone (r_h) and the outer radius of the particle ring (r_o) are set to 1 cm and 3 cm, respectively. Ambient conditions outside the high-pressure region are initialized at a pressure (p_0) of 1 atm and temperature (T_0) of 300 K. Simulations are performed for six distinct initial conditions, as summarized in Table 1, where pH, T_h and d_p denote the pressure and temperature of the explosive source, and the particle diameter, respectively. Case A serves as the baseline scenario, while Cases B, C-D, and E-F form comparative series designed to examine the effects of shock reflection, particle diameter, and explosion energy, in turn. The confined case uses a 1.5 m × 1.5 m domain (one-quarter of a 3 m × 3 m laboratory-scale chamber). The open-space domain is extended to 2.4 m × 2.4 m to avoid particle loss at boundaries and maintain the same total number of particles in both configurations. Thus, the key difference is the presence or absence of reflected shocks. Prior to the blast wave reaching $x = 1.5$ m or $y = 1.5$ m, both cases yield identical results. For Case A, the initial conditions within the high-pressure region correspond to the constant-volume explosion combustion limit of Trinitrotoluene (TNT) with a density of 1.654 g/cm^3 [4].

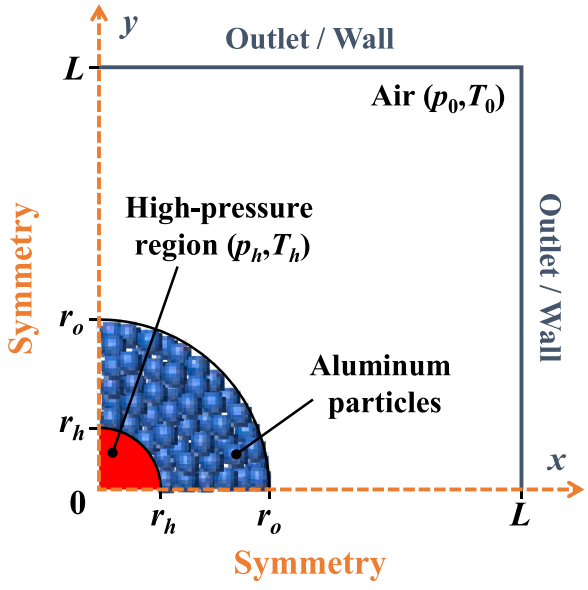


Fig. 2. Schematic of the physical model for numerical simulation.

Table 1
Physical parameters for all simulated cases.

Case	p_h (GPa)	T_h (K)	d_p (μm)	L (m)	Confined / Open
A	9	2780	5	1.5	Confined
B	9	2780	5	2.4	Open
C	9	2780	2	1.5	Confined
D	9	2780	8	1.5	Confined
E	5	2780	5	1.5	Confined
F	1	2780	5	1.5	Confined

2.2. Numerical methods

To simulate gas-particle reactive flows spanning from dense to dilute regimes during the explosive dispersal and combustion of a particle ring, this study employs a four-way coupled compressible multi-phase particle-in-cell (CMP-PIC) method [24,26] within an Eulerian-Lagrangian framework. The continuous gas phase is governed by the unsteady compressible multi-species reactive Navier-Stokes equations, which account for volume fraction effects and interphase coupling, and are solved on a uniform Eulerian grid. The dispersed particle phase is tracked individually in Lagrangian coordinates. Inter-particle collisions are resolved using the coarse-grained discrete element method. For Al particles, with a Young's modulus of 68 GPa and a Poisson's ratio of 0.33, the stiffness coefficients for particles of different diameters are calculated using Hertzian contact theory and the coarse-grained criterion [27]. Considering that micron-aluminum particles lie in the transition regime for heat and mass transfer, a Knudsen-number-dependent regime-based formulation [28] is employed to calculate the interphase heat transfer rate, while a hybrid combustion model [25] that accounts for both diffusion reaction and surface kinetic oxidation is adopted to calculate the reaction rate. This model has been widely adopted in previous studies [19,29,30]. The thermodynamic properties of the gas mixture are obtained by invoking the CHEMKIN package [31], which computes seven-coefficient polynomial fits as functions of gas temperature. Meanwhile, the transport properties are evaluated by using the TRANSPORT package [32].

The governing equations of the two-phase reactive flow are solved with the following numerical algorithms. For the gas phase, a fifth-order weighted essentially non-oscillatory (WENO) scheme [33] is used for primitive variable reconstruction; the Harten-Lax-van Leer (HLL) Riemann solver [34] is applied to calculate the convective fluxes; and a fourth-order central difference scheme is employed to discretize the viscous fluxes. The particle-phase equations are discretized via the Velocity-Verlet algorithm [35]. Two-phase coupling is achieved through a cubic spline interpolation function [36]. Temporal integration is performed with the third-order Runge-Kutta scheme. Detailed formulations of the governing equations and numerical algorithms can be found in our previous work [24]. To accelerate the simulation, an adaptive computational domain expansion method is developed. Further details are provided in Section A of the Supplementary Material. Grid independence verification is presented in Section B of the Supplementary Material.

2.3. Validation of numerical model and method

2.3.1. Shock-dense particle cloud interaction

The CMP-PIC method has been validated against benchmark experiments in our previous studies on shock-driven dense particle problems, including shock interaction with dense particle curtains [24], shock-induced dust lifting [37], and inert particle dispersion driven by a central explosion [26]. The simulation results show good agreement with experimental data, demonstrating that this method can accurately capture the interaction process between blast waves and dense particle clouds.

2.3.2. Ignition delay time of micron-aluminum particle

To assess the reliability of the numerical model in simulating particle combustion and interphase heat transfer, we simulate the heating, ignition, and combustion of a single Al particle. The setup follows the experimental conditions reported by Roberts et al. [38]. The particle has an initial diameter of 21.7 μm and an initial surface temperature of 300 K. The ambient gas is a mixture of 99% O_2 and 1% N_2. A high-temperature, high-pressure gas environment is achieved using a reflected shock tube, with the post-shock pressure reaching 17 atm.

Fig. 3(a) shows the particle temperature evolution at a gas temperature of 2400 K. The particle ignition process comprises three distinct stages. The first is the preheating stage, during which the particle is heated from its initial temperature (T_{p0}) to the melting point of Al (T_m) via interphase heat transfer, with a duration denoted as t_{Pre}. The second stage is melting, where the particle absorbs heat for phase transition while maintaining a constant temperature; the time required for complete melting is recorded as t_{Mel}. Subsequently, the particle temperature rises again until reaching the critical ignition temperature (T_{ig}), indicating successful ignition of the particle; this period is defined as the fast oxidation stage, with duration t_{Fo}. Based on experimental data [39], the critical ignition temperature of an Al particle is fitted as: $T_{ig} = e^{0.087\ln(d_p)+7.28}$, where d_p is the particle diameter in μm. The total time from T_{p0} to T_{ig} is defined as the ignition delay time ($t_{IDT} = t_{Pre} + t_{Mel} + t_{Fo}$). Fig. 3(b) compares the calculated results with the experimental data from [38] over an ambient temperature range of 2400–3000 K. The predictions obtained with the present numerical model and method show good agreement with the measured values.

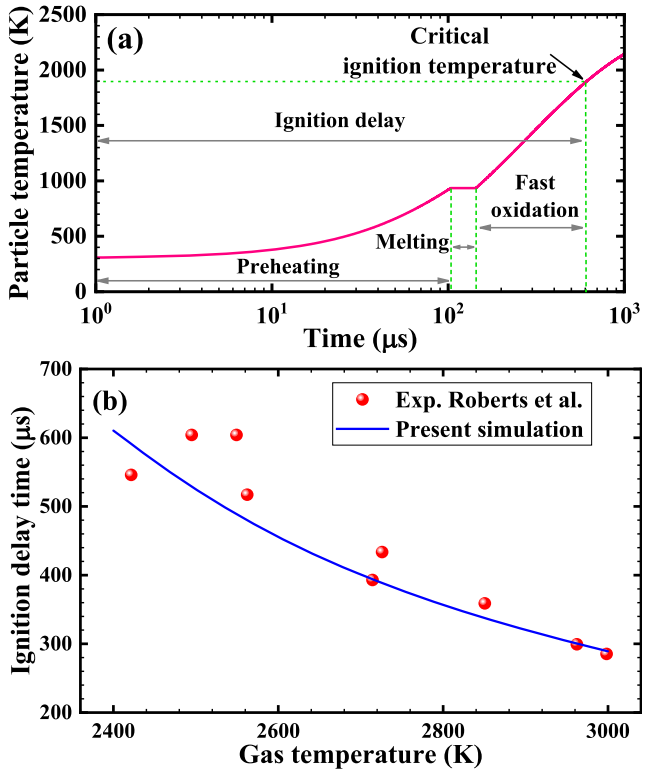


Fig. 3. Single-particle ignition simulation results: (a) temporal evolution of particle temperature; (b) comparison of calculated ignition delay time with experimental data.

2.3.3. Ignition and combustion of al particle clouds induced by shock impact

Recently, we applied this method to conduct numerical simulations of shock-induced two-phase detonation combustion in air-Al particle clouds [29]. Under various particle concentration and size conditions, the obtained detonation velocities, peak pressures, and cellular sizes all agree well with experimental observations. This further validates the reliability of the numerical method in simulating the combustion behavior of Al particle clouds under shock loading.

3. Results and discussion

3.1. Dispersion and combustion of Al particles in a confined space

3.1.1. Particle cloud dispersion and evolution

In the initial stage, the gas-particle mixture within the ring is extremely dense, with α_p well above 1%. As a result, Al particle combustion is oxygen-limited. Under the impact of the central blast wave, the particles gradually disperse and mix with air, forming a two-phase combustible mixture whose equivalence ratio progressively enters the flammable range. Therefore, this subsection first presents the particle dispersion and evolution process.

Fig. 4 shows the temporal evolution of particle distribution and cloud morphology. Driven by the divergent blast wave, the particle ring expands outward. Initially, the particle cloud remains clustered, and no significant dispersion is observed (Fig. 4(a)). By 0.28 ms, particles at the inner edge begin to disperse, while those at the outer edge stay clustered. Notably, initial perturbations appear at the outer edge at this instant (Fig. 4(b)). As the leading transmitted shock propagates outward, the cloud disperses under the combined effects of the post-shock flow and interparticle collisions. Concurrently, the initial perturbations grow into numerous unstable finger-like particle jets (Fig. 4(c)). These jets subsequently break up, and the particles further disperse and mix with air, supplying oxidizer for particle combustion. By 0.84 ms, the cloud retains an overall annular shape (Fig. 4(d)). At 0.98 ms, the leading shock reaches the domain boundary and collides with the wall, generating a reflected shock that travels back toward the cloud and exerts a secondary impact on it. Owing to their high inertia, the particles continue moving radially outward; those at the outer edge change direction upon collision with the upper and right walls. By 1.4 ms, most particles lie behind the reflected shock (Fig. 4(f)). At 1.82 ms, some particles have experienced two reflected shocks (Fig. 4(g)). Thereafter, the shock waves undergo multiple reflections within the confined space, leading to thorough dispersion of the cloud.

Fig. 5(a) shows the radial evolution of the average particle volume fraction over time. Initially, the thickness of the particle ring changes very little. Around 0.3 ms, due to the velocity difference between the inner and outer edges of the ring, its thickness gradually increases and the particle cloud progressively disperses. As the central pressure decays and the reflected shock interacts with the cloud, particles at the inner edge begin to move inward toward the center at 1.25 ms. By 2.4 ms, particles are distributed throughout the entire domain.

Fig. 5(b) displays the temporal evolution of the various forces acting on all particles, enabling granular dynamics analysis. During the initial shock impact, the particle ring experiences peak shock intensity. Driven by strong gas forces, inner-ring particles move outward, causing interparticle collisions and a sharp rise in collision force. Although gas forces and collision force peak nearly simultaneously, the collision force

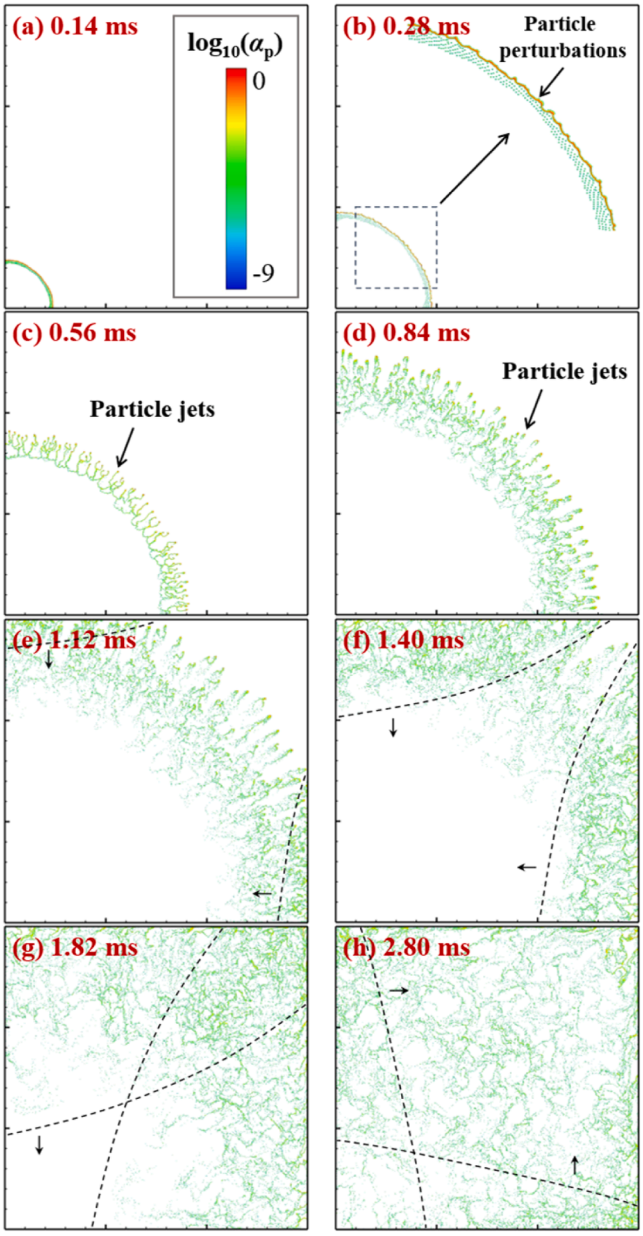


Fig. 4. Temporal evolution of particle volume fraction contours for Case A. The domain size is 1.5 m × 1.5 m.

is significantly larger, indicating that particle cloud dynamics in this stage are dominated by inter-particle collisions.

As the ring expands outward, inter-particle spacing increases, causing rapid decay of the collision force. After the shock penetrates the ring, the drag and pressure gradient forces also decay quickly, with the pressure gradient force decaying most rapidly. The drag force decreases to a certain level and then remains relatively stable; after 0.1 ms, it is consistently an order of magnitude larger than the pressure gradient force. By about 0.65 ms, the collision force falls below the drag force and continues to decline. As the reflected shock impacts the cloud and the outer particles collide with the walls, all forces exhibit a slight step increase. During brief intervals, the collision force can again exceed the drag force. After this second impact, inter-particle spacing further increases and collision frequency decreases; thus, after 1.4 ms, the collision force becomes significantly smaller than the drag force, which persists at a certain level. Consequently, in the middle to late stages, the evolution of the particle cloud is primarily dominated by the drag force.

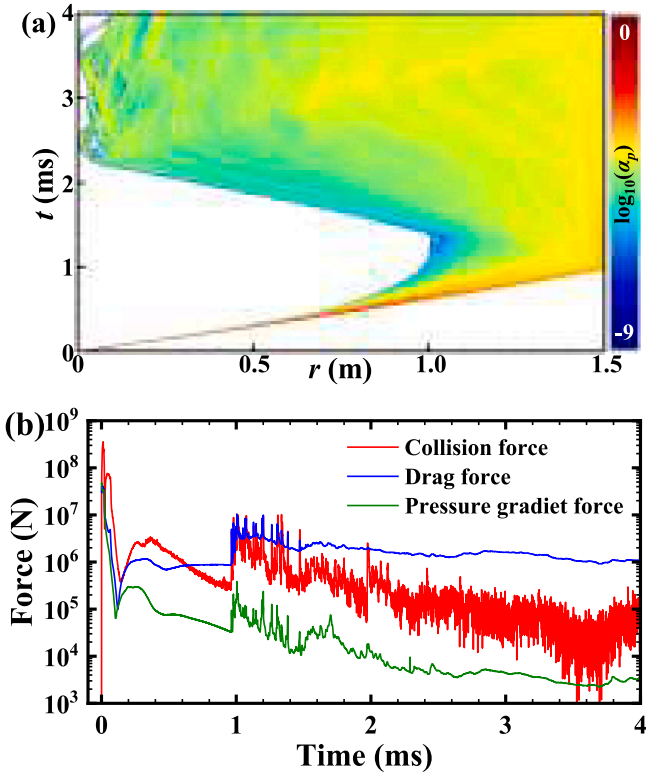


Fig. 5. (a) Spatiotemporal evolution of particle volume fraction. (b) Temporal evolution of collision, drag, and pressure gradient forces acting on the entire particle cloud.

3.1.2. *Combustion of Al particles in the cloud*

Comparison of Figs. 4(a-b) and 6(a-b) indicates that the blast wave transmits rapidly through the particle ring, subsequently inducing a gradual dispersion of the particle cluster. Another significant effect of the blast wave is compressing and heating the gas. At the instant of Fig. 6 (b), the post-shock gas temperature reaches ~1750 K, which greatly exceeds the melting point of Al particles (933 K). Intense convective heat transfer from the shock-heated gas to the particle cluster continuously raises the particle temperature, initiating reactions in a small fraction of the particles. This phenomenon is corroborated by the oxygen mass fraction (Y_{O2}) distribution in Fig. 7. At the outer edge of the ring, where particles are more densely concentrated, oxygen consumption is more significant. At this stage, most particles remain highly packed with a volume fraction exceeding 40%, and their combustion is primarily limited by the oxygen content in the gas. As shown in Figs. 7(a) and (b), the oxygen concentration inside the clusters is nearly zero.

During the dynamic dispersion of the particle cloud, Al particles continuously mix with the surrounding air, allowing ambient oxygen to infiltrate the cluster interstices. Meanwhile, the combined effect of high-temperature gas gradually induces combustion of Al particles behind the shock. As shown in Figs. 6(c) and 7(c), partial Al combustion has begun, and the fresh oxygen within the cloud is being continuously consumed. The heat released from the particle reactions elevates the gas temperature around the particles to a level significantly higher than that of the shock-heated gas. As dispersion proceeds, both the spatial extent of the particle cloud and the combustion zone expand considerably. Furthermore, Fig. 6(d) clearly demonstrates a marked increase in gas temperature at the location of the cloud, where multiple high-temperature ignition kernels reaching 4000 K are present. Nevertheless, the combustion rate for most particles remains relatively low, constrained by two factors. First, the shock intensity continuously decays. Compared to gaseous fuels, metallic particle fuels exhibit a longer ignition delay time; thus, the post-shock gas temperature is insufficient to promptly propel

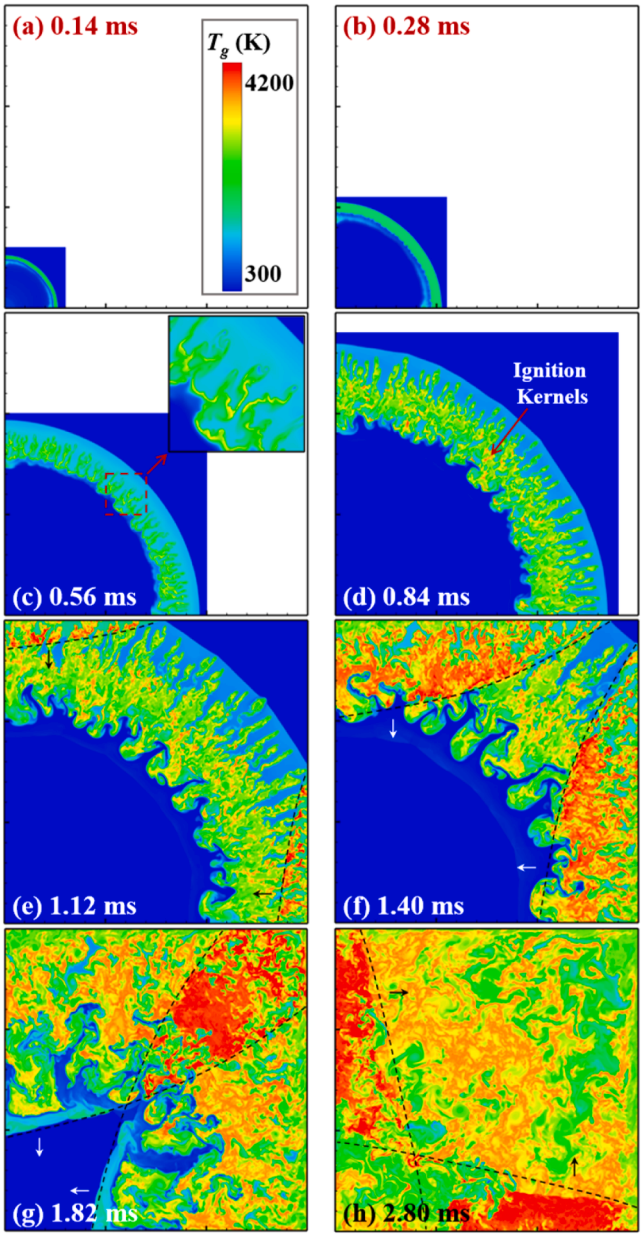


Fig. 6. Temporal evolution of gas temperature contours for Case A. The domain size is 1.5 m × 1.5 m.

Al particles into a rapid combustion regime. Second, many particles remain heavily clustered, and their combustion rate is still largely limited by oxygen availability, as indicated in Fig. 7(d).

By 1.12 ms, the leading shock has propagated to and interacted with the upper and right solid wall boundaries, generating reflected shock waves that travel in the opposite direction (Fig. 6(e)). The reflected shocks exert a secondary impact on the Al particle cloud, further raising the post-wave gas temperature and pressure beyond the levels induced by the incident shock. This enhanced effect provides the Al particles with greater ignition energy, significantly shortening their ignition delay time. Moreover, the impact of the reflected shocks and collisions of the particle cloud with the walls further promoted particle dispersion. Consequently, the gas temperature contours in Figs. 6(e-f) distinctly show markedly higher temperatures in the region affected by the reflected shocks, with localized areas exceeding 4200 K. This outcome stems from the above factors significantly accelerating the combustion rate and energy release of the Al particles.

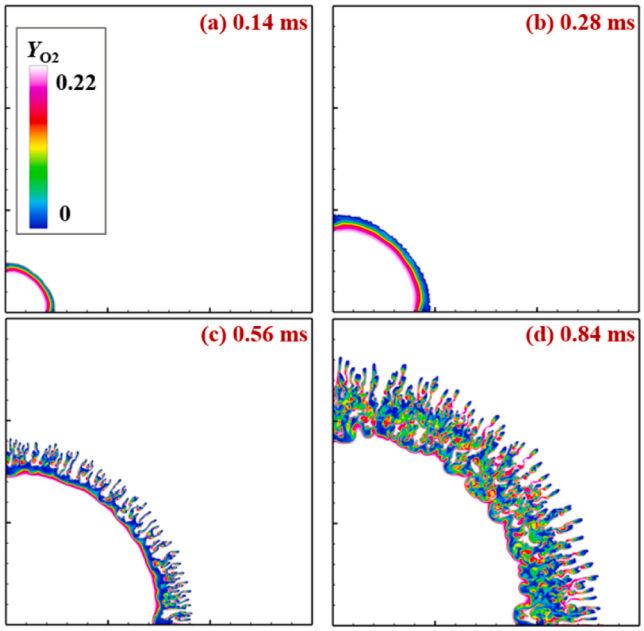


Fig. 7. Contours of oxygen mass fraction at different times for Case A. The domain size is 1.5 m × 1.5 m.

At 1.82 ms, the two reflected shocks intersect near the diagonal of the domain (Fig. 6(g)). Their collision raises the gas pressure and temperature, further increasing the combustion rate of particles subjected to successive impacts from both reflected shocks, thereby boosting the heat release. The flow field temperature in these regions is notably higher than elsewhere, appearing deep red in the figure. As the two reflected shocks propagate, the area swept by both expands continuously, resulting in a significant increase in high-temperature zones above 4000 K. When the reflected shocks reach the left and bottom walls, a second shock-wall interaction occurs, generating new reflected shocks that further heat the gas and induce particle combustion through shock impact. At this stage, the combustion rate of the particle clusters subjected to three successive shock interactions increases further, elevating both the heat release from combustion and the gas temperature. In Fig. 6(h), the temperature contour in the region swept by three shocks shows the deepest red, with the local flow field temperature reaching up to 4500 K. Note that the area near the origin experiences the effect of four reflected shocks; however, due to the relatively low Al particle count there, its temperature does not exceed that of the triple-shocked region with a higher particle concentration. Subsequent evolution continues in this cyclic manner until the particles are fully combusted or the available oxygen is depleted.

3.2. *Mechanisms of initial disturbance formation and jet growth*

This subsection analyzes the mechanisms underlying the two key stages of particle dispersion identified in Section 3.1: the formation of initial perturbations (i.e., jet initiation) and subsequent jet growth. We first discuss the emergence of initial perturbations. Fig. 8 shows the radial velocity (v_r) distributions of particle clusters during 7–224 μs, with the lower-left curve displaying the v_r of the outer-edge particles. Fig. 9 presents the radial forces acting on these outer-edge particles. Scatter plots of radial forces on all particles are provided in Section C of the Supplementary Material.

At 7 μs, a compression wave forms in the granular medium driven by the blast wave, with the compression front (CF) leading the pressure front (PF). By 14 μs, the CF reaches the outer edge of the cluster. The heterogeneity of collision processes transmitted radially induces significant variations in collision forces (Fig. 9(a)), producing highly uneven

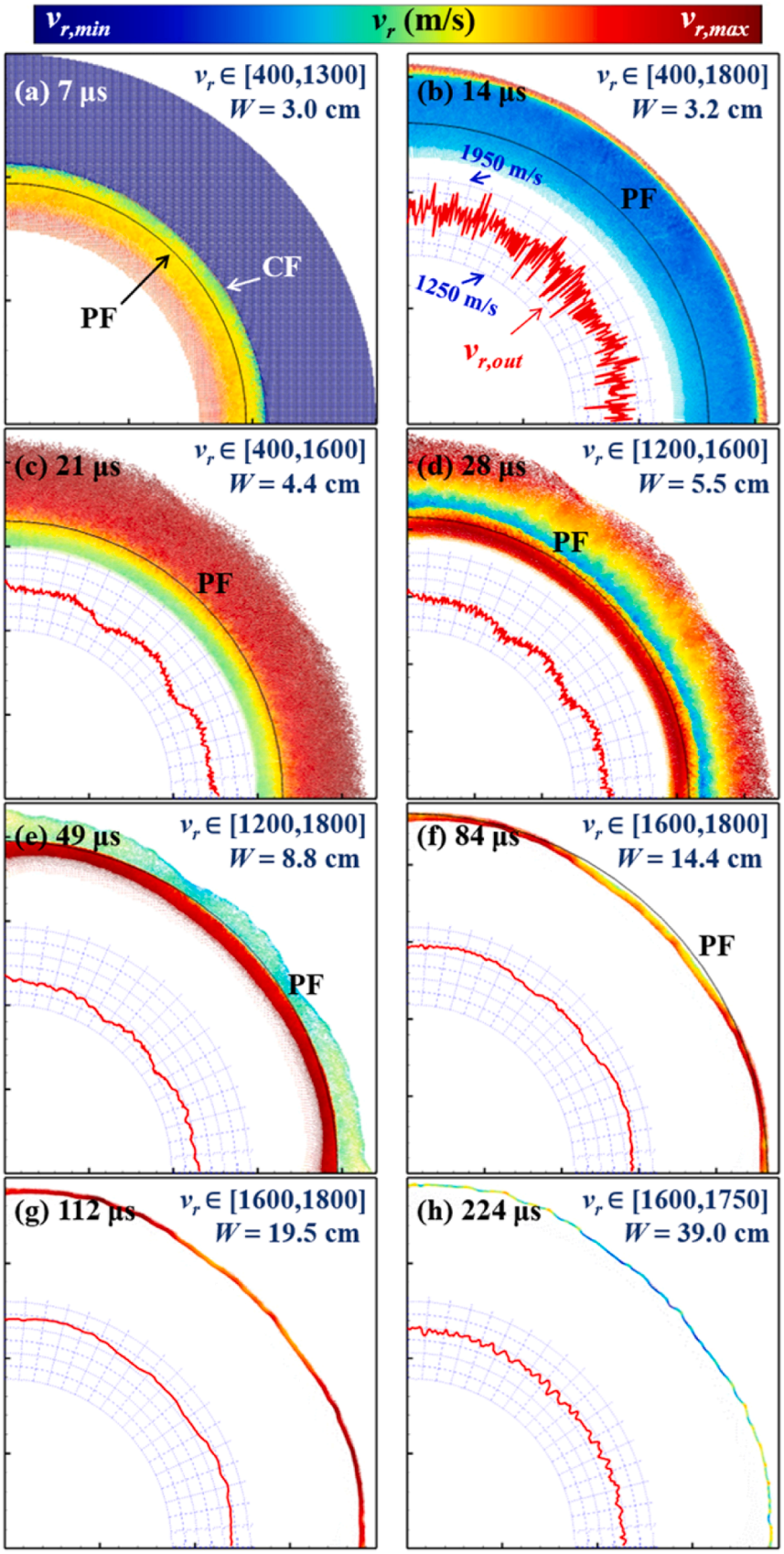


Fig. 8. Radial velocity distributions of the particle cloud and outer-edge particles at different times.

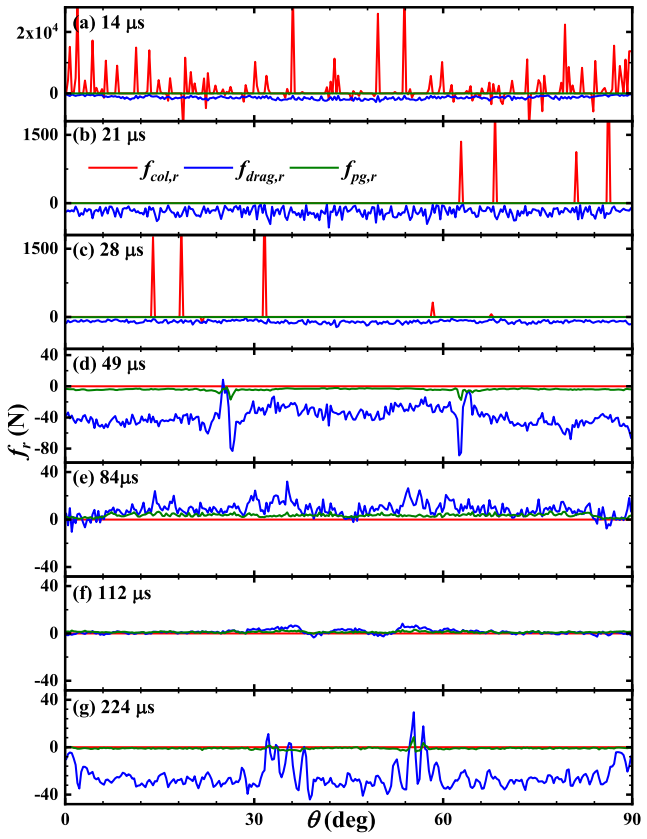


Fig. 9. Circumferential distribution of forces acting on outer-edge particles.

outer-edge velocities (Fig. 8(b)). Many particles move faster than the gas, generating a negative radial drag force directed toward the origin. Velocity differences among particles further cause varying drag forces. Inner particles are driven outward by the shock wave and gas flow but constrained by outer particles; their gained momentum is transferred to the outer particles via collisional forces. By 21 μs, particles outside the PF move substantially faster than those inside, with pronounced heterogeneity emerging at the cloud periphery (Fig. 8(c)). As the post-PF gas velocity increases and the shock drives the particles, a secondary compression wave forms within the cluster, with the CF and PF nearly coinciding (Fig. 8(d)). Particles behind the CF re-accelerate, while those outside the CF decelerate under drag retardation; this contrast is evident in Fig. 8(e). The heterogeneous force distribution on the outer-edge particles produces significant peripheral perturbations (Fig. 8(c-e)). However, under the action of relatively circumferentially uniform pressure waves and secondary compression waves, these perturbations are mitigated (Fig. 8f). Notably, by this time substantial heterogeneities in particle position and volume fraction already exist within the cluster, leading to heterogeneous interphase momentum exchange.

Figs. 10(a-c) show the radial velocities of particles at various positions after perturbation formation. Outer-edge particles consistently exhibit greater v_r than the inner ones. This velocity difference transforms the initial perturbation into particle jets and drives the continuous elongation of jet length (Figs. 4(c), (d)). As the leading shock weakens and the overall cluster velocity declines, velocity differences between particles at different radial positions increase. Fig. 10(d-f) presents the radial distribution of average radial forces on particles at identical radial positions. After the PF penetrates the cloud, both drag and pressure gradient forces become negative, with drag dominant. The most salient feature is that the outer-edge particles generally experience weaker drag than inner particles, thereby producing the velocity profiles in Figs. 10 (a-c). This arises because outer-edge particles, being closer to the leading shock, encounter higher local gas velocities and smaller interphase

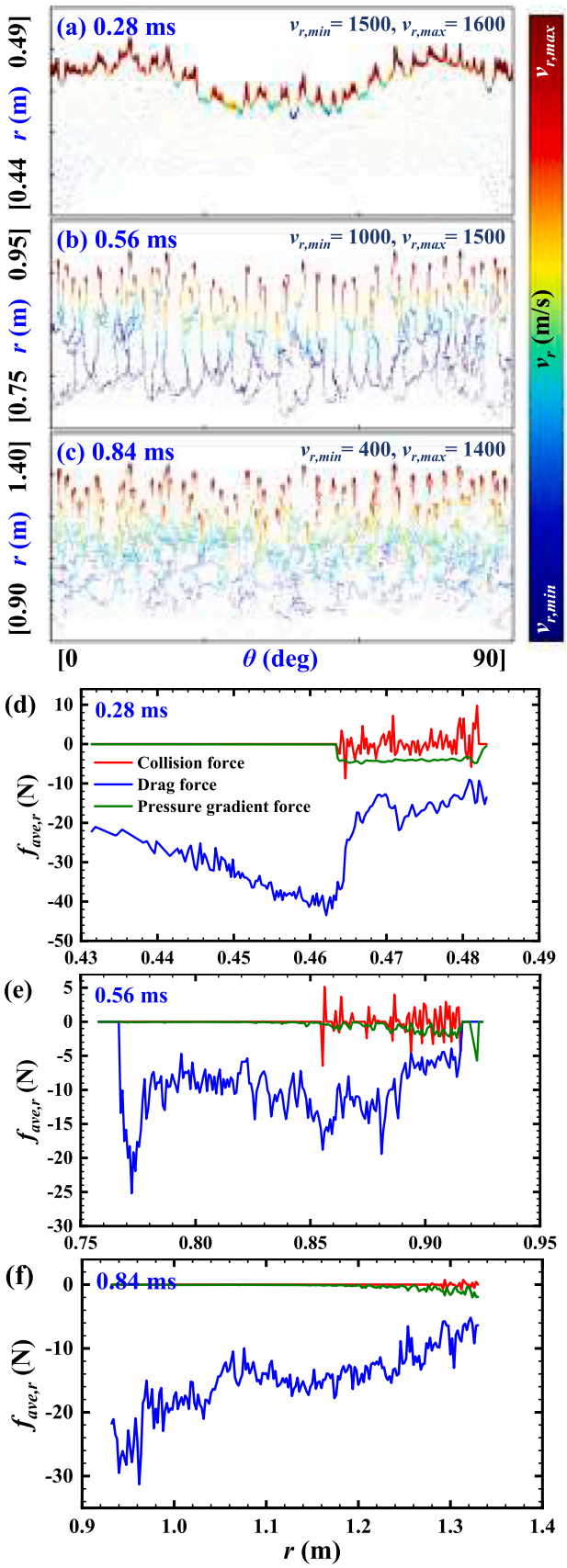


Fig. 10. (a-c) Radial velocity of particles at different positions, and (d-f) average radial force on particles at different radial positions.

velocity differences, resulting in weaker drag retardation than inner particles. Additionally, increasing velocity differences among particles induces interparticle collisions; inner particles exert outward radial collision forces on outer particles (i.e., inner particles push outer particles while outer particles obstruct inner particles). These two factors jointly cause outer particles to decelerate more slowly and maintain higher velocities.

Fig. 11 presents the gas-phase vorticity distributions at two typical instants, corresponding to the onset of particle perturbations and the development of particle jets, to analyze the gas-particle instability and its effects. According to Koneru et al. [17], vorticity generation in two-phase flows arises from three mechanisms: the baroclinic torque, the torque due to the coupling force, and the misalignment between the volume-fraction-weighted gas density and the coupling force. As shown in Fig. 11(a), substantial vorticity is deposited near the outer edge of the cloud during the perturbation-initiation stage. Since this region lies outside the cloud, the vorticity is attributed primarily to the baroclinic torque. This vorticity is organized into alternating positive and negative vortex pairs, which induce azimuthal momentum on the particles. Specifically, a vortex pair with positive vorticity on the left side and negative vorticity on the right drives particles toward the center of the vortex pair, forming outward protrusions of the particle cloud; the opposite configuration leads to inward concavities, as depicted in Fig. 11 (a2). These features indicate that the particle cloud exhibits the characteristic imprint of the Richtmyer-Meshkov instability. At the instant shown in Fig. 11(b), well-defined particle jets have formed, and vortex-pair structures persist but are now largely confined within the particle cloud, suggesting that interphase interactions dominate vorticity generation at this stage. Particles on the left side of the jet diffuse azimuthally leftward, inducing positive vorticity, while those on the right diffuse rightward, inducing negative vorticity. The spatial configuration of these vortex pairs is consistent with the observations reported by Koneru et al. [17].

To clarify the role of collisions in perturbation onset and development, idealized collisionless simulations were performed. Fig. 12 compares results with and without collisions. At 0.28 ms, the collisional case exhibits significant outer-edge perturbations, whereas the collisionless case maintains a relatively smooth arc-shaped morphology, despite developing internal volume fraction inhomogeneities. By 0.56 ms, particle jets also appear in the collisionless case, but are shorter than those in the collisional case. Thus, gradual volume fraction non-uniformization and the uneven action of drag forces during dispersion are the key factors inducing particle perturbations and jets, while collision forces accelerate initial perturbation formation and jet growth. This occurs because non-uniform radial collisions enhance particle distribution heterogeneity and increase the radial velocities of outer particles. Furthermore, collisions impart a certain circumferential velocity component and reduce the radial velocity, causing slightly lower radial expansion rates in the collisional case than in the collisionless case. Simulation results for cases with different stiffness coefficients are also provided in the Supplementary Material to demonstrate the influence of collision force magnitude. The results show that a larger stiffness coefficient leads to earlier perturbation formation and longer jets.

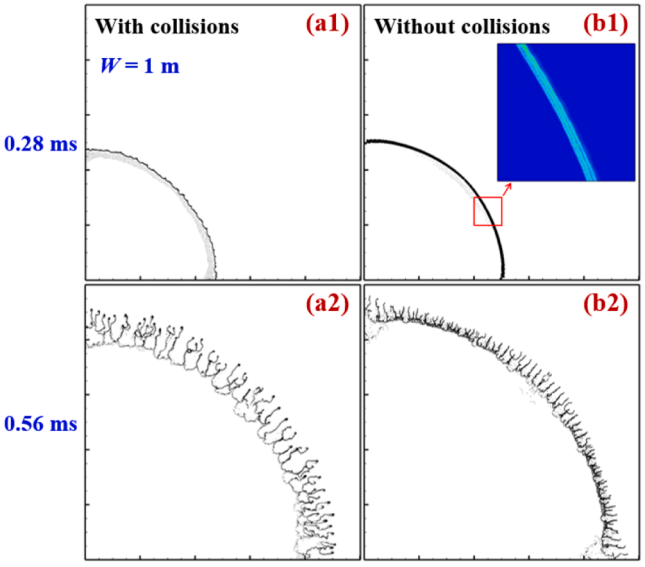


Fig. 12. Particle distributions in the collisional and collisionless cases at different times.

3.3. Effects of reflected shock waves

The following compares results from confined and open-space scenarios to analyze the effects of spatial confinement and reflected shock waves on the combustion of an Al particle cluster. More detailed results for the open-space scenario are presented in the Supplementary Material. Fig. 13 displays the spatiotemporal contours of the radially averaged gas-phase temperature, clearly visualizing the evolution of wave patterns. In the confined case, the initial blast wave impacts the cloud while compressing and heating the surrounding gas. As the particles and gas are mixed to a certain degree (t ≈ 0.6 ms), the cloud enters the combustion stage, marked by a moderate temperature rise in Region I of

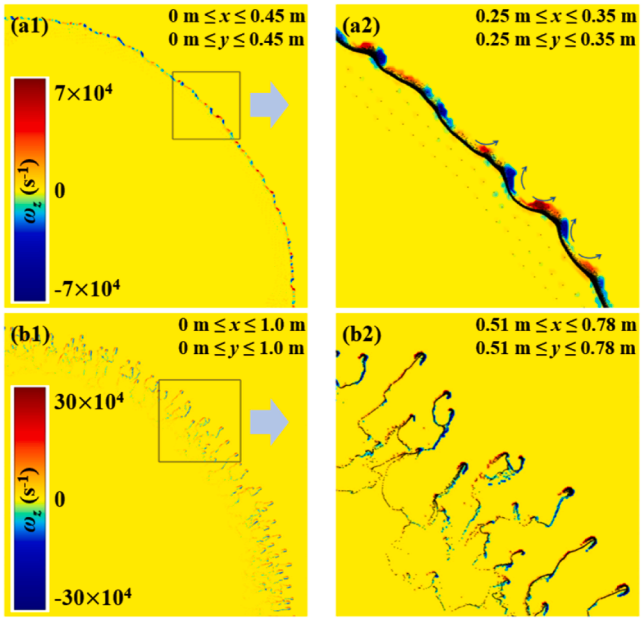


Fig. 11. Gas vorticity contours at (a) 0.252 ms and (b) 0.588 ms; left column shows enlarged views of the local region.

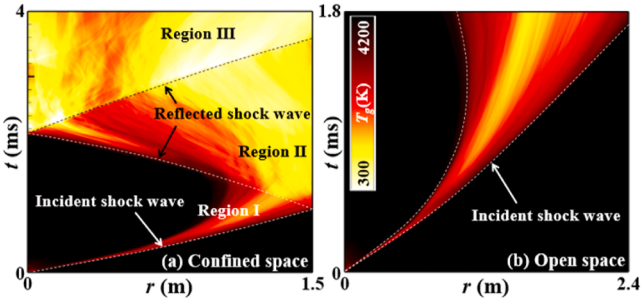


Fig. 13. Position-time diagrams of gas temperature in the radial direction for the (a) confined and (b) open scenarios.

Fig. 13(a). At ~0.96 ms, the incident shock collides with the wall, generating a primary reflected shock that further heats the gas and exerts a secondary impact on the cloud. Following this, both the particle combustion rate and gas temperature continue to rise (Region II). At ~2.15 ms, a secondary reflection occurs, further elevating the flow temperature (Region III). A comparison across regions distinctly shows that the gas temperature increases sequentially from Region I to Region II and then to Region III. Fig. 13(b) presents the results of the open-space scenario. Under the impact of the blast wave, the cloud exhibits a sustained dispersive motion pattern. In the absence of wall confinement, both the incident shock and the cloud outer boundary expand freely outward. As particle dispersion increases, the local oxygen concentration within the cloud rises. Consequently, the combustion zone expands steadily, developing a fan-shaped morphology with elevated gas temperatures persisting at its center. Comparison of the two configurations reveals that the confined case produces a combustion zone with both higher temperatures and greater spatial extent.

Fig. 14(a) shows the combustion rate of all Al particles and the normalized unburned Al mass (scaled by the initial Al particle mass, $m_{p,0}$) versus time under both boundary conditions. During the extremely short period at the explosion onset, the high-energy shock causes the Al particles to rapidly consume the oxygen originally present in the ring, yielding a remarkably high burning rate. Once the oxygen in the ring is fully depleted, the combustion rate drops sharply to near zero and persists at this level for a period. As particles continue to disperse, the combustion process is gradually reinitiated, causing the combustion rate to rise again alongside a continuous decrease in Al mass.

For the confined-space scenario, when the reflected shock begins to interact with the cloud, the Al combustion rate increases significantly, accompanied by a rapid decrease in Al mass. Afterward, as the reflected shock weakens and the Al particle diameter decreases, the combustion rate decreases accordingly. Due to the limited simulation duration, a portion of Al remains unburned at the end of the calculation. Under open-space conditions, the combustion rate, after peaking, maintains a relatively stable level for a period before declining. By 1.76 ms, particles at the outer edge begin to exit the computational domain. A direct comparison in Fig. 14(a) shows that during the interval of 1.0–1.4 ms, the interaction with the reflected shock in the confined scenario significantly enhances the particle combustion rate relative to the open-space case, with the peak rate increasing by approximately 110%. However, due to the reduced oxygen concentration and rapid particle depletion, the combustion rate in the confined space declines to a level comparable to that in the open-space case by approximately 1.6 ms.

To evaluate the primary factors contributing to the enhancement of combustion by the reflected shock, Fig. 15 presents the averaged oxygen density and gas temperature at all particle positions. Upon the reflected shock, the oxygen density increases markedly, accompanied by a relatively slower rise in gas temperature. Thus, the increase in oxygen content within the cloud induced by the reflected shock is primarily

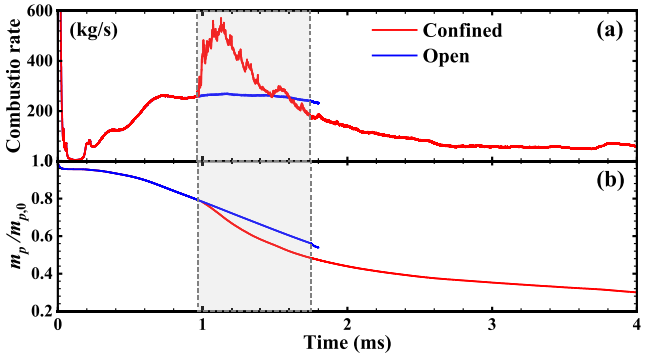


Fig. 14. (a) Al particle burning rate and (b) remaining Al mass normalized by the initial total mass for Cases A and B.

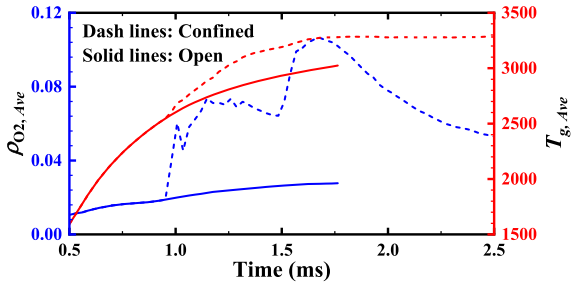


Fig. 15. Averaged oxygen density and gas temperature around the particle.

responsible for the enhanced combustion rate, followed by the shock-induced gas heating. The post-shock oxygen density rise within the cloud is mainly attributed to the shock compression, which increases gas density and hence oxygen supply, and to the improved gas-particle mixing resulting from the shock-promoted particle dispersion.

3.4. Effects of particle diameter

This subsection presents simulations examining the influence of initial Al particle diameter (2 μm and 8 μm) on cloud dispersion and combustion. Fig. 16 shows the spatiotemporal evolution of particle volume fraction and gas temperature along the radial direction for both cases. Overall, the dispersion patterns and combustion development exhibit essentially similar characteristics across the two particle sizes. However, differences in particle diameter and mass alter the forces acting on the particles, as well as the heat transfer and reaction rates, introducing distinct evolutionary features at different stages.

According to the particle volume fraction contours in the left column of Fig. 16, the outer edges of the particle clouds reach the upper and lower boundaries of the computational domain at 1.02 ms and 0.93 ms for d_p = 2 μm and 8 μm, respectively. This indicates that particle diameter exerts a certain degree of influence on the overall dispersion rate of the cluster, thereby shifting the combustion progression. Comparison of the gas temperature distributions shows that temperatures in Regions I and II are higher for the 8 μm case than for the 2 μm case. To detail these differences, Fig. 17 shows the particle cloud and gas temperature distributions at 0.7 ms. The cloud with larger particles exhibits superior early-stage dispersion, manifested as greater cloud thickness and more dispersed individual particle jet structures. Enhanced particle

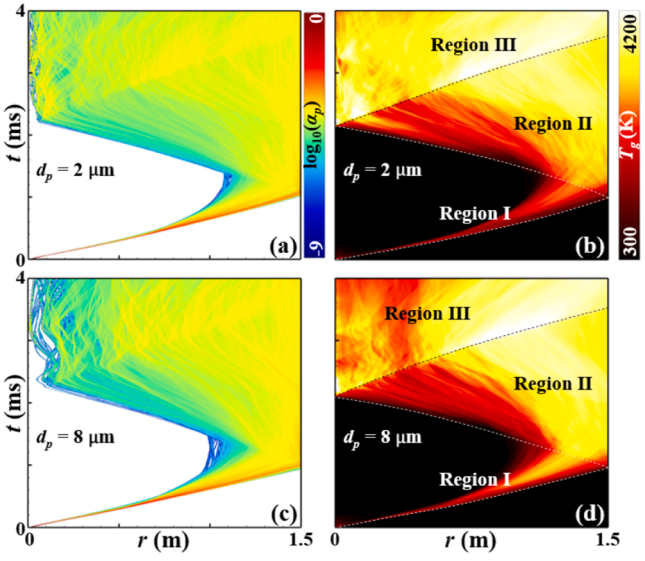


Fig. 16. Spatiotemporal diagrams of particle volume fraction and gas temperature in the radial direction for particle diameters of 2 μm and 8 μm.

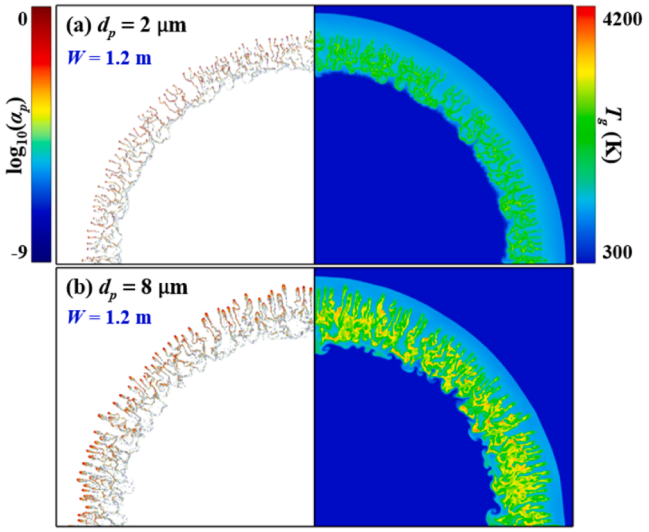


Fig. 17. Results at 0.7 ms for (a) d_p = 2 μm and (b) d_p = 8 μm. The left half shows the particle distribution, and the right half displays the gas temperature contour.

dispersion promotes higher combustion rates. As shown in the temperature contours in Fig. 17, numerous particles in the 8 μm case have already entered the rapid combustion regime, yielding significantly higher gas temperatures compared to the 2 μm case, where no discernible ignition kernels appear. Subsequently, under the influence of the reflected shock wave and the inward pressure gradient, the particle cloud converges toward the origin. Comparisons show that the 2 μm particles disperse more rapidly and thoroughly across the entire domain, while for the 8 μm particles only a small fraction returns to the origin vicinity between 2.4 and 3.0 ms. Moreover, the temperature contours indicate that Region III in the small-particle case is notably hotter than that in the large-particle case.

The following analysis examines the effects of particle size from a particle dynamics perspective. Fig. 18 shows the time histories of collision and drag forces acting on particles for the three particle sizes considered. Initially, fine particles experience significantly larger drag force. Under identical flow conditions, the drag force formulation implies that finer particles undergo higher acceleration. Because the total mass of the cloud is the same in all cases, the 2 μm cloud experiences the maximum drag force. In contrast, collision force is related to individual particle mass, leading to a higher initial peak for larger particles (Fig. 18 (b)). As the shock propagates through the cloud and particles disperse, both forces decay to low values in all three cases. Among them, fine particles experience slightly higher drag forces, while large particles endure greater collision forces. When the reflected shock interacts with the cloud and the cloud collides with the wall, both forces rise rapidly. Before the reflected shock arrives, drag forces continuously pull particles radially outward, causing internal particles to accumulate toward the cloud periphery. Because fine particles experience greater drag forces while the collision forces that promote dispersion are weaker, their dispersion efficiency is lower at this stage, as shown in Fig. 17. Once the reflected shock impinges on the cloud, the drag force direction reverses and dominates particle motion. Consequently, at later stages fine particles disperse more effectively, burn faster, and yield higher temperatures in Region III.

Fig. 19 presents the total combustion rate and remaining Al mass for all particles in the three cases. The results for the 5 μm and 8 μm cases are quite similar, while both differ significantly from the 2 μm case. During the early to middle stages, despite their smaller size, the 2 μm particles display a relatively low combustion rate. As analyzed above, this stems from the poor dispersion of finer particles at this stage, wherein combustion is constrained by oxygen availability within the cloud. In the middle to late stages, however, the dispersion performance of the 2 μm particles improves significantly. After 1.15 ms, the burning rate for the 2 μm case consistently exceeds that of the other two cases.

3.5. *Effects of initial central pressure*

To further investigate the influence of blast wave intensity, simulations are performed at initial central pressures of 1 GPa and 5 GPa, with all other parameters held constant (Fig. 20). In the 1 GPa case, the weakest initial shock significantly slows cloud dispersion: the cloud front reaches the domain boundary at approximately 2.7 ms, versus 1.2 ms for the 5 GPa case and 0.98 ms for the 9 GPa case. The evolution of the 5 GPa case resembles that of the baseline scenario, while the 1 GPa case exhibits distinct characteristics. As shown in Fig. 20(a), for the 1 GPa case, inner particles begin to flow back toward the origin as early as 1.4 ms, whereas the incident shock and outer particles reach the boundary at 2.2 ms and 2.7 ms, respectively. This indicates that the particle flow-back is not caused by the reflected shock. The primary reason is the rapid pressure drop at the center in the 1 GPa case, which generates an inwardly propagating expansion wave that drives the inner particles back toward the origin. After reaching the center, the expansion wave reflects and partially pushes particles away from the center. Subsequently, it oscillates between the origin and the inner edge of the cloud. The temperature contour shows a rise in gas temperature following the reflected expansion wave. Meanwhile, the reflected shock propagates inward from the outer edge, and the two waves collide at approximately 3.4 ms. Due to its low intensity, the reflected shock

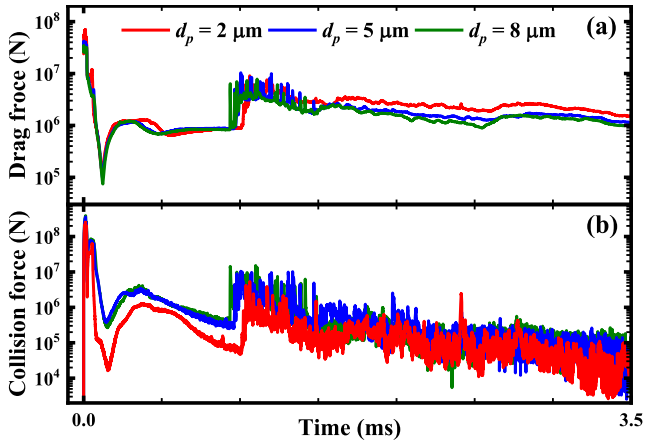


Fig. 18. Variation of (a) drag forces and (b) collision forces acting on the entire particle cloud with time under different particle diameter conditions.

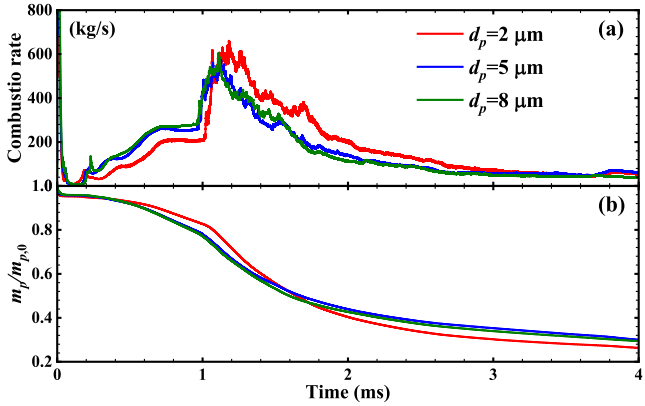


Fig. 19. Variation of (a) combustion rate and (b) remaining Al mass of the particle cloud with time under different particle diameter conditions.

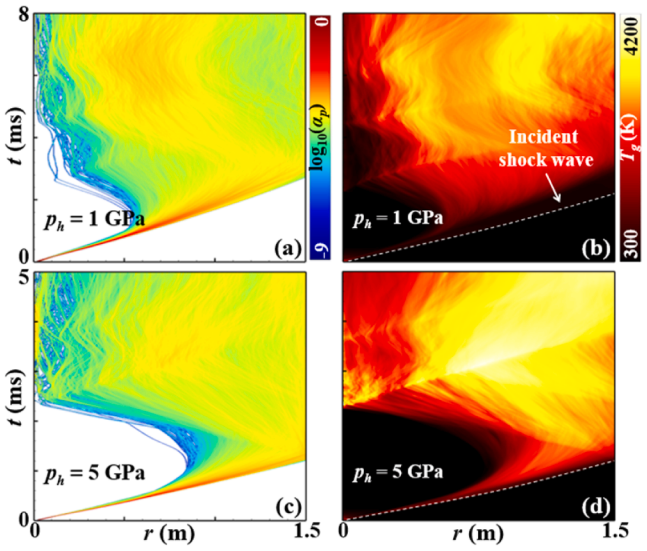


Fig. 20. Radial spatiotemporal diagrams of particle volume fraction and gas temperature for pH = 1 GPa and 5 GPa.

cannot penetrate the entire particle cloud or undergo multiple reflections across the domain, as occurs in the baseline case. The combined effects of the inward expansion wave and the external reflected shock cause particle combustion to occur primarily in the central radial region.

Fig. 21 shows the temporal variations of drag and collision forces acting on the cloud for the three central pressure conditions. Reducing the initial central pressure significantly decreases both forces. In the 1 GPa case, the reflected shock is extremely weak. Consequently, the drag force does not exhibit the step-like increase observed in the other two cases, but instead rises slowly from 2.6 ms onward. The collision force is likewise minimally affected by the reflected shock. Fig. 22 plots the total combustion rate and remaining Al mass over time. As shown in Fig. 22 (a), a higher initial central pressure yields an earlier and higher peak combustion rate. As combustion proceeds, the Al particle diameter decreases rapidly, causing a sharp decline in the overall combustion rate. For the 5 GPa and 9 GPa cases, the combustion rates converge to nearly identical values in the later stage, indicating that increasing the central pressure beyond a certain level primarily affects the early- to mid-stage combustion characteristics. In the 1 GPa case, the time to reach the peak burning rate is more than twice that of the 5 GPa case, and the peak value is considerably lower. To quantitatively evaluate the effect of central pressure on particle combustion efficiency, results at $t = 3.0$ ms are compared. The Al mass consumption rate in the 9 GPa case is 3.6% and 58.8% higher than in the 5 GPa and 1 GPa cases, respectively, while the 5 GPa case shows a 57.0% improvement over the 1 GPa case.

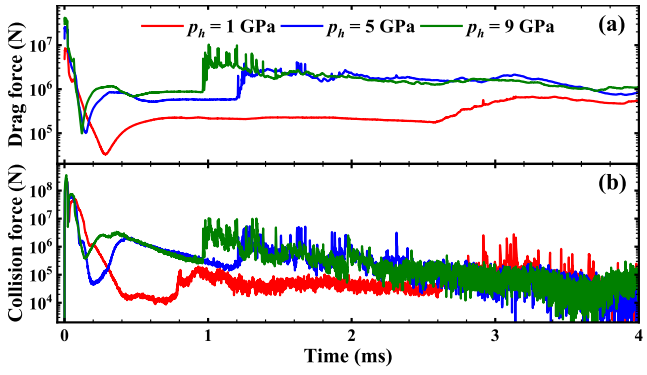


Fig. 21. Evolution of (a) drag force and (b) collision force on the particle cloud under different initial central pressures.

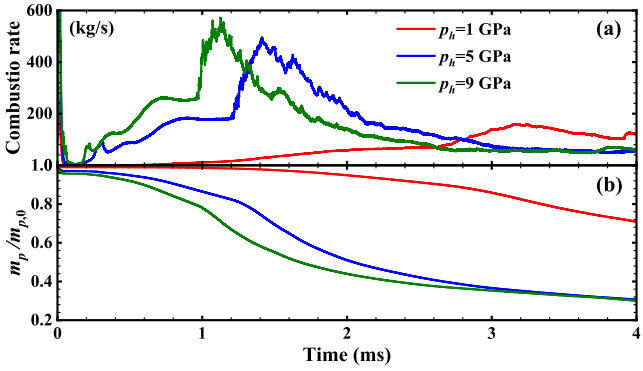


Fig. 22. Variation of (a) combustion rate and (b) remaining Al mass of the particle cloud with time under initial central pressure conditions.

Therefore, increasing the central pressure within a certain range can significantly enhance particle combustion efficiency, but further increases beyond this range yield substantially diminishing benefits.

4. Conclusions

This study employed a four-way coupled Eulerian-Lagrangian method to simulate the dispersion and combustion of a dense Al particle layer driven by a central explosion. The effects of spatial confinement, particle diameter, and initial explosion energy are systematically investigated.

The dispersion dynamics and combustion process are first analyzed. Driven by the divergent blast wave, initial perturbations appear at the outer edge of the ring and develop into finger-like particle jets that enhance gas-particle mixing and facilitate combustion. As dispersion proceeds, particle motion transitions from early collision-dominated to later drag-dominated behavior. Regarding combustion characteristics, particles begin to react after the blast wave transmits through the ring. Initially, most particles remain in a highly packed state (volume fraction >40%), and combustion is oxygen-limited. With further dispersion, ambient oxygen gradually infiltrates the cloud, forming multiple ignition kernels reaching 4000 K. Subsequent interaction with reflected shock waves further elevates gas temperature and enhances particle combustion, with regions subjected to successive shock impacts exhibiting temperatures as high as 4500 K.

The mechanisms governing perturbation formation and jet growth are then elucidated. Analysis of early-stage particle evolution and radial velocity distributions shows that two successive compression waves form within the cloud, accompanied by intense collisions. The first compression wave induces outer-edge perturbations via heterogeneous collision forces; these perturbations are largely smoothed by subsequent waves, yet leave a non-uniform particle distribution. This inhomogeneity leads to uneven drag on outer-edge particles, seeding perturbations at the cloud periphery. Vorticity diagnostics further show that, during this stage, baroclinic torque deposits substantial vorticity near the cloud outer edge, forming alternating vortex pairs that induce azimuthal particle motion and create protrusions or concavities, indicating a Richtmyer-Meshkov instability imprint. During jet development, outer-edge particles, being closer to the wave front, encounter higher gas velocities and thus experience weaker drag retardation, sustaining higher radial velocities that drive jet elongation. The vortex-pair structures persist but are largely confined within the cloud, indicating that interphase interactions dominate vorticity generation at this stage. Furthermore, comparison between collisional and collisionless simulations confirms that inter-particle collisions accelerate both perturbation initiation and jet development.

Reflected shock waves in confined spaces exert secondary impacts on the particle cloud, significantly enhancing the combustion rate. The

peak burning rate increases by approximately 110% compared to open-space conditions. The enhancement arises primarily from increased oxygen concentration via shock compression and improved gas-particle mixing, and secondarily from elevated gas temperature. Spatiotemporal temperature diagrams demonstrate that the gas temperature increases sequentially through regions subjected to successive shock reflections, indicating that spatial confinement can be leveraged to improve energy release efficiency in composite charges.

Particle diameter exhibits a dual effect on dispersion and combustion. Smaller particles (2 μm) experience greater drag forces but weaker collisional dispersion in the early stage, resulting in poorer dispersion and lower combustion rates. After reflected shock interaction, however, they disperse more effectively and burn faster, achieving higher gas temperatures in the later stage compared to larger particles (8 μm). The combustion rate curves show that after 1.15 ms, the burning rate for the 2 μm case consistently exceeds that of the 5 μm and 8 μm cases, indicating that particle size optimization must consider the entire temporal evolution of the dispersion-combustion process.

Increasing initial pressure advances combustion onset and raises the peak burning rate. However, the enhancement effect saturates beyond 5 GPa, with the combustion rates of the 5 GPa and 9 GPa cases converging to similar values in the later stage. The Al mass consumption at 3.0 ms in the 9 GPa case is 3.6% and 58.8% higher than that in the 5 GPa and 1 GPa cases, respectively, suggesting that an optimal pressure range exists for maximizing energy output. For the 1 GPa case, the weaker shock and the resulting inwardly propagating expansion wave lead to distinctly different combustion characteristics, with particle combustion occurring primarily in the central radial region.

In future work, we will consider more realistic particle ignition and combustion models, further investigate more realistic three-dimensional spherical configurations, incorporate polydisperse particle size distributions, and examine the effects of turbulence on dispersion and combustion dynamics.

Supplementary material

Supplementary material associated with this article can be found, in the online version.

CRediT authorship contribution statement

Jiahui He: Writing – original draft, Validation, Methodology, Investigation, Data curation, Conceptualization. **Jianling Li:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition. **Baoqing Meng:** Writing – review & editing, Methodology. **Baolin Tian:** Supervision, Methodology. **Chuanyu Pan:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2026.115297.

References

[1] V.R. Schuetz, D.S. Stewart, Eigenvalue detonation: particle size effects in aluminized explosives, Combust. Flame 279 (2025) 114256.
[2] Q. Pontalier, J. Loiseau, S. Goroshin, F. Zhang, D.L. Frost, Blast enhancement from metalized explosives, Shock Waves. 31 (3) (2021) 203–230.
[3] J.M. Peuker, H. Krier, N. Glumac, Particle size and gas environment effects on blast and overpressure enhancement in aluminized explosives, Proc. Combust. Inst. 34 (2) (2013) 2205–2212.
[4] J.W. Posey, B. Roque, S. Guhathakurta, R.W. Houim, Mechanisms of prompt and delayed ignition and combustion of explosively dispersed aluminum powder, Phys. Fluids 33 (11) (2021) 113308.
[5] W.A. Trzcinski, L. Maiz, Thermobaric and enhanced blast explosives-properties and testing methods, Propellants, Explos., Pyrotech. 40 (2015) 632–644.
[6] E. Fedina, C. Fureby, Analysis of heat-release during TNT/aluminum afterburning by means of numerical simulations, Proc. Combust. Inst. 36 (2017) 2841–2848.
[7] X. Duan, X. Guo, Q. Jiao, J. Zhang, Q. Zhang, Effects of Al/O on pressure properties of confined explosion from aluminized explosives, Def. Technol. 13 (6) (2017) 428–433.
[8] Y. Tang, W. Dong, X. Zou, B. Shi, N. Wang, Ignition and combustion of a dense powder jet of micron-sized aluminum particles in hot gas, Proc. Combust. Inst. 39 (3) (2023) 3625–3636.
[9] V. Bagheri, M.H. Keshavarz, A. Mousaviazar, The effect of active aluminum content on the detonation performance of aluminized explosives, J. Energ. Mater. 41 (2) (2023) 144–156.
[10] J. Nie, R. Kan, Q. Jiao, Q. Wang, X. Guo, S. Yan, Studies on aluminum powder combustion in detonation environment, Chin. Phys. B 31 (4) (2022) 044703.
[11] Z. Wang, F. Chen, P. Liu, Y. Zhou, C. Xiao, Shock activation theory for aluminum nano-particles outside high explosives, Combust. Flame 272 (2025) 113882.
[12] D.L. Frost, Y. Gregoire, O. Patel, S. Goroshin, F. Zhang, Particle jet formation during explosive dispersal of solid particles, Phys. Fluids 24 (2012) 091109.
[13] J. Li, W. Li, J. Yu, W. Xiao, H. Xu, Blast performance of layered charges enveloped by aluminum powder/rubber composites in confined spaces, Def. Technol. 18 (4) (2022) 583–592.
[14] C. Xiao, Z. Wang, F. Chen, Y. Xu, X. Jiao, The efficient activation by adding energetic metallic particles outside high explosives: a new direction for composite explosives, J. Energ. Mater. (2023) 1–18.
[15] V. Rodriguez, R. Saurel, G. Jourdan, L. Houas, Solid-particle jet formation under shock-wave acceleration, Phys. Rev. E 88 (2013) 063011.
[16] K. Xue, L. Miu, J. Li, C. Bai, B. Tian, Explosive dispersal of granular media, J. Fluid. Mech. 959 (2023) A17.
[17] R.B. Koneru, B. Rollin, B. Durant, F. Ouellet, S. Balachandar, A numerical study of particle jetting in a dense particle bed driven by an air-blast, Phys. Fluids 32 (2020) 093301.
[18] A. Chiapolino, R. Saurel, Numerical investigations of two-phase finger-like instabilities, Comput. Fluids. 206 (2020) 104585.
[19] D.L. Frost, Heterogeneous/particle-laden blast waves, Shock Waves. 28 (2018) 439–449.
[20] D.L. Frost, F. Zhang, S.B. Murray, S. McCahan, Critical conditions for ignition of metal particles in a condensed explosive, in Proceedings of the 12th International Detonation Symposium.
[21] K. Balakrishnan, A.L. Kuhl, J.B. Bell, V.E. Beckner, An empirical model for the ignition of explosively dispersed aluminum particle clouds, Shock Waves. 22 (2012) 591–603.
[22] K. Shimura, A. Matsuo, Using an extended CFD-DEM for the two-dimensional simulation of shock-induced layered coal-dust combustion in a narrow channel, Proc. Combust. Inst. 37 (2019) 3677–3684.
[23] K. Balakrishnan, S. Menon, On turbulent chemical explosions into dilute aluminum particle clouds, Combust. Theor. Model. 14 (4) (2010) 583–617.
[24] J. He, B. Meng, H. Li, B. Tian, J. Li, Heterogeneous detonation in gas-particle mixtures with full pattern flows: numerical model, method, and verification, Comput. Phys. Commun. 316 (2025) 109797.
[25] F. Zhang, K. Gerrard, Reaction mechanism of aluminum-particle-air detonation, J. Propuls. Power. 25 (2009) 845–858.
[26] B. Tian, J. Zeng, B. Meng, Q. Chen, X. Guo, K. Xue, Compressible multiphase particle-in-cell method (CMP-PIC) for full pattern flows of gas-particle system, J. Comput. Phys. 418 (2020) 109602.
[27] J. Zeng, B. Meng, C. Jun, B. Tian, Coarse-grained numerical simulation for compressible fluid–particle two-phase flows, Phys. Fluids 35 (2023) 053315.
[28] X. Zou, N. Wang, J. Wang, C. Wang, L. Han, B. Shi, Numerical study on the characteristics of a nano-aluminum dust-air jet flame, Aerosp. Sci. Technol. 121 (2022) 107304.
[29] J. He, B. Meng, B. Tian, H. Li, J. Li, Numerical study of two-phase detonation in dilute aluminum particle-air mixtures using eulerian-lagrangian approach, Combust. Flame 275 (2025) 114110.
[30] P. Zhang, Y. Xu, H. Zhang, Z. Chen, Unsteady dynamics and mode transitions in hybrid hydrogen-aluminum detonations, Combust. Flame 281 (2025) 114438.
[31] R.J. Kee, F.M. Rupley, J.A. Miller, Chemkin-II: a Fortran chemical kinetics package for the analysis of gas phase chemical kinetics. Technical Report, 1989.
[32] R.J. Kee, G. Dixon-Lewis, J. Warnatz, A Fortran Computer Code Package for the Evaluation of gas-phase, Multicomponent Transport Properties, Sandia Report Sand, 1986.
[33] X. Liu, S. Osher, Weighted essentially non-oscillatory schemes, J. Comput. Phys. 115 (1994) 200–212.
[34] E.F. Toro, Riemann Solvers and Numerical Methods for Fluid Dynamics: A Practical Introduction, Springer-Verlag, Berlin, Heidelberg, 1999.

[35] A.M. Tartakovsky, A. Panchenko, Pairwise force smoothed particle hydrodynamics model for multiphase flow: surface tension and contact line dynamics, J. Comput. Phys. 305 (2016) 1119–1146.
[36] J.P. Morris, P.J. Fox, Y. Zhu, Modeling low reynolds number incompressible flows using SPH, J. Comput. Phys. 136 (1997) 214–226.
[37] J. Li, J. He, B. Meng, B. Tian, Investigation of dust lifting by a moving shock wave based on compressible multiphase particle-in-cell method, Phys. Fluids 34 (2022) 103316.
[38] T.A. Roberts, R.L. Burton, Ignition and combustion of aluminum/magnesium alloy particles in O_2 at high pressures, Combust. Flame 92 (1993) 125–143.
[39] Y. Huang, G.A. Risha, V. Yang, R.A. Yetter, Effect of particle size on combustion of aluminum particle dust in air, Combust. Flame 156 (1) (2009) 5–13.